SYNTHETIC CONTROL

Generalized synthetic controls (gsynth)

- Method distinction:
    - Unifies original synthetic control with interactive fixed effects model, DiD is a special case.
    - "Generalized" because it extends original synthetic control to the case of multiple treated units and variable treatment periods.
    - The general steps are:
        - 1. Use untreated observations to estimate factors in a latent factor model: common factors that vary with time and influence outcomes. The number of factors can be chosen via cross-validation.
        - 2. Use treated units pre-treated data to fit their factor loadings (these vary by unit but not over time).
        - 3. Use the now fully fit factor model to predict control outcomes post-treatment for treated units
    - Because of the parametric setup, it's more likely that a good synthetic control can't be constructed here - important to validate **[this may be why confidence intervals are sometimes so large]**
    - Covariates can be included for learning the factor model, potentially improving precision
    - I don't see why the matrix completion method wouldn't work in theory for one treated unit, but Yiqing Xu commented this on Github, and for sure it is not currently implemented for the package
- Standard error methods:
    - Parametric bootstrap is used for confidence intervals (non-parametric option, but the documentation recommends parametric when the number of treated units is small)
        - In each iteration, randomly select one control unit as if it were treated instead
        - Resample the rest of the control group with replacement
        - Apply GSC to the new sample, getting a vector of prediction errors e^P for the placebo treated unit
        - Apply GSC to original data, getting vector of prediction errors e for each control unit
        - For a number of iterations:
            - Start with the GSC original model predictions for control units, with "control" set that samples w/ replacement an added error from e and a "treated" set that samples w/ replacement an added error from e^P. This is a fully placebo sample - no real treatment effects.
            - Apply GSC method to S and get an ATT estimate, add this to the GSC original method estimate
        - Use these to create permutation distribution for p-values, CIs

Micro synthetic controls (microsynth)

- Method distinction:
    - Allows multiple outcomes to be used at once - assumes unobserved confounders are common to all outcomes.
    - But, our current use case is one outcome, one treated unit: this package just boils down to conventional synthetic control in that case, possibly with covariates used in the matching/alignment algo.
- Standard error methods:
    - They offer three methods: linearization, jackknife, and permutation. They say linearization is "best practice" in general, but not trustworthy with a single treated unit. Similarly, they advise not to use jackknife in this case (see their JSS paper p. 23).
    - They advise only to use p-values from the permutation approach, but not confidence intervals (package won't return confidence intervals from permutation in our case) (p. 25).

Augmented synthetic controls (augsynth)

- Method distinction:
    - Uses an outcome model to estimate bias due to imperfect pre-treatment fit, de-biases the original synthetic control method.
    - Covariates can be included in both the original synthetic controls model and the outcome model for estimating bias.
    - Doubly robust - if either the synthetic control reweighting or the outcome correction model works well, low bias
- Standard error methods:
    - Conformal option (refer to Chernozhukov et al. 2021 - I found this clearer than the aug synth paper)
        1. For a given sharp null hypothesis \tau = \tau_0, edit observation of treated unit Y_1T - \tau_0 as the "new" treatment data (we usually just consider tau_0 = 0 so no change).
        2. Apply the augsynth method to the adjusted data set to get adjusted weights and treated unit residuals under "control" in post-treatment periods, then aggregate residuals for the test stat (bigger → rejection)
        3. Compute p-value: what proportion of times is the treated outcome residual from step 2 larger than residuals when we permute time periods?
        4. Compute confidence intervals: repeat this for a fine grid of \tau_0 values, the set of \tau_0 values in which p > \alpha (where we cannot reject the sharp null) is our pointwise confidence interval.
        5. Practical notes: this only appears to be implemented for some of the bias correction methods, does not apply to: elastic net (EN), random forest (RF) or neural networks (seq2seq)
- Jackknife and jackknife+

- Mentioned in the github for the package, but not in the paper. They note it requires additional assumptions but they give no theoretical results or explicit assumptions on how they perform in this setting for synthetic control.

- Jackknife+ is a variation on jackknife (leave-one-out predictions) where you just use the LOO prediction as the point estimate, not the estimate from the full data model. Based on the documentation, jackknife+ seems to be implemented to leave out one *time period*, while jackknife leaves out one *unit*.

- Jackknife method only returns standard errors in the current package implementation. To get confidence intervals and p-values, our code uses a (potentially inaccurate) normal approximation, following what their inner code does for the plot summary function.